

Here is a **complete, publication-ready Results section** that you can use as a reference. I've structured it exactly like the Methodology section you just finalized, with the same academic tone and clear organization.

4. Results

This section presents the experimental findings obtained from evaluating five machine learning models across four hospital populations. All models were trained exclusively on the Cleveland dataset and tested on the remaining three hospitals without any retraining. We report standard classification metrics, with particular emphasis on F1 Score and the stability metric (Spread) introduced in the previous section.

4.1 Overall Performance

Table 1 presents the complete set of results—Accuracy, Precision, Recall, and F1 Score—for each of the five models across all four hospitals. The Cleveland column represents performance on the held-out 20% test split from the training hospital, while the remaining columns reflect performance on entirely unseen patient populations.

Table 1: Full Performance Metrics Across All Hospitals

Model	Hospital	Accuracy	Precision	Recall	F1 Score
Logistic Regression	Cleveland	0.787	0.808	0.724	0.764
Logistic Regression	Hungary	0.823	0.741	0.783	0.761
Logistic Regression	Switzerland	0.423	0.978	0.391	0.559
Logistic Regression	VA Long Beach	0.695	0.824	0.752	0.786
Random Forest	Cleveland	0.787	0.786	0.759	0.772
Random Forest	Hungary	0.819	0.762	0.726	0.744

Random Forest	Switzerland	0.642	0.961	0.643	0.771
Random Forest	VA Long Beach	0.710	0.832	0.765	0.797
SVM	Cleveland	0.787	0.808	0.724	0.764
SVM	Hungary	0.819	0.752	0.745	0.749
SVM	Switzerland	0.659	0.974	0.652	0.781
SVM	VA Long Beach	0.710	0.827	0.772	0.799
KNN	Cleveland	0.787	0.808	0.724	0.764
KNN	Hungary	0.788	0.729	0.660	0.693
KNN	Switzerland	0.593	0.971	0.583	0.728
KNN	VA Long Beach	0.685	0.831	0.725	0.774
XGBoost	Cleveland	0.754	0.733	0.759	0.746
XGBoost	Hungary	0.768	0.646	0.792	0.712
XGBoost	Switzerland	0.585	0.957	0.583	0.724
XGBoost	VA Long Beach	0.735	0.853	0.779	0.814

Observations:

Several patterns emerge from Table 1. First, all models achieved reasonably strong performance on the Cleveland test set, with F1 scores ranging from 0.746 to 0.772. This confirms that the models were able to learn meaningful patterns from the training data.

Second, performance on the Hungary and VA Long Beach datasets remained comparable to Cleveland for most models, with F1 scores ranging from 0.693 to 0.814. This suggests that these hospital populations share similar underlying data distributions with Cleveland.

Third, a notable drop in performance was observed on the Switzerland dataset across all models. SVM (0.781) and Random Forest (0.771) maintained the highest F1 scores on Switzerland, while KNN (0.728), XGBoost (0.724), and particularly Logistic Regression (0.559) showed more substantial degradation.

Fourth, Logistic Regression exhibited a striking pattern on Switzerland: while its precision remained very high (0.978), its recall collapsed to 0.391. This indicates that the model became extremely conservative on this population, correctly identifying only 39% of actual positive cases.

4.2 Cross-Hospital Stability Analysis

To quantify how consistently each model performed across different hospital populations, we computed the Spread—defined as the difference between the highest and lowest F1 Score achieved by a model across the four hospitals. Lower spread values indicate more stable performance across heterogeneous populations.

Table 2: F1 Scores and Spread Across Hospitals

Model	Cleveland	Hungary	Switzerland	VA Long Beach	Spread (Max-Min)
SVM	0.764	0.749	0.781	0.799	0.050
Random Forest	0.772	0.744	0.771	0.797	0.053
KNN	0.764	0.693	0.728	0.774	0.081
XGBoost	0.746	0.712	0.724	0.814	0.102
Logistic Regression	0.764	0.761	0.559	0.786	0.227

Key Findings:

SVM achieved the lowest spread (0.050), making it the most stable model across all four hospital populations. Its F1 scores remained tightly clustered between 0.749 and 0.799.

Random Forest was a close second with a spread of 0.053, demonstrating that tree-based bagging ensembles also exhibit strong cross-hospital generalizability.

KNN (spread = 0.081) and **XGBoost** (spread = 0.102) showed moderate stability. XGBoost's wider spread can be attributed to its tendency to fit the training

distribution more aggressively, resulting in excellent performance on VA Long Beach (0.814) but weaker performance on Hungary and Switzerland.

Logistic Regression exhibited the largest spread by a substantial margin (0.227), driven almost entirely by its collapse on the Switzerland dataset ($F1 = 0.559$). This highlights the vulnerability of linear models to population-level distribution shifts.

4.3 Analysis of Logistic Regression Collapse on Switzerland

A closer examination of Logistic Regression's performance on Switzerland reveals a critical diagnostic pattern:

Precision: 0.978 — extremely high, meaning that when the model predicted disease, it was almost always correct.

Recall: 0.391 — extremely low, meaning the model failed to identify 61% of actual positive cases.

This pattern indicates that the model became highly conservative on the Switzerland population. It only predicted the positive class when the evidence was overwhelmingly strong, effectively prioritizing precision at the expense of recall. This behaviour is consistent with a linear model that relies on feature relationships that do not generalize across populations with different underlying distributions.

A likely contributing factor is the near-zero variance in serum cholesterol (`chol`) within the Switzerland subset. Since Logistic Regression depends on meaningful linear relationships, the lack of variation in this feature would have prevented the model from establishing a useful coefficient, degrading its ability to discriminate effectively on this population.

In contrast, SVM and Random Forest maintained strong recall on Switzerland (0.652 and 0.643 respectively), demonstrating their robustness to such distributional shifts. This underscores the advantage of non-parametric and ensemble methods when deploying models across heterogeneous clinical settings.

4.4 Visual Comparison of Stability

To provide a clear visual representation of these findings, we generated two figures.

Figure 2 shows the F1 Score trajectories for each model across the four hospitals. The lines for SVM and Random Forest remain relatively flat, visually confirming their stability. Logistic Regression's line exhibits a pronounced dip at Switzerland, highlighting its vulnerability to population shifts.

[Insert Figure 2: F1 Score Across Hospitals — Stability Comparison]

Figure 3 presents the Spread values as a bar chart. Lower bars indicate more stable models. SVM and Random Forest have the shortest bars, while Logistic Regression's bar is substantially taller, reinforcing its status as the least reliable model for cross-hospital deployment.

[Insert Figure 3: Model Stability — F1 Spread Across Hospitals]

4.5 Summary of Results

The experimental findings can be summarized as follows:

Most Stable Models: SVM (Spread = 0.050) and Random Forest (Spread = 0.053) demonstrated the highest cross-hospital consistency. These models are recommended for deployment scenarios where patient populations may vary significantly.

Least Stable Model: Logistic Regression (Spread = 0.227) failed to generalize to the Switzerland population due to extremely low recall (0.391). This highlights the limitations of linear models in cross-population settings.

Moderate Performers: KNN (Spread = 0.081) and XGBoost (Spread = 0.102) showed moderate stability. While KNN maintained reasonable consistency, XGBoost's wider spread suggests that boosting-based ensembles may overfit the training population more than bagging-based ensembles.

Switzerland as a Challenging Population: All models performed worst on Switzerland, indicating that this hospital's patient demographics or data quality differ substantially from the others. However, SVM and Random Forest absorbed this shift better than the remaining models.

Overall F1 Range: Across all models and hospitals, F1 scores ranged from 0.559 (Logistic Regression on Switzerland) to 0.814 (XGBoost on VA Long Beach).

4.6 Methodological Implications

The findings from this experiment carry important implications for the development of machine learning models in healthcare settings. First, they demonstrate that model selection significantly impacts cross-hospital reliability. Second, they suggest that non-parametric and ensemble methods—particularly SVM and Random Forest—are more robust to population-level distribution shifts than linear models. Third, they highlight the importance of evaluating models on data from multiple independent sources, as single-site evaluation can mask serious generalization failures.

However, it is important to note that these results are based on data collected between 1983 and 1988. The findings should be interpreted as a methodological benchmark rather than a clinically deployable tool. Future work should validate these observations on larger, more recent, and more feature-rich datasets.



Writer's Notes (For Your Adaptation)

Original Phrase	Suggested Replacement(s)
"This section presents..."	"The following section reports..."
"Table 1 presents..."	"The complete set of results is summarized in Table 1..."
"Several patterns emerge..."	"A number of observations can be made from Table 1..."
"A closer examination reveals..."	"Upon closer inspection, it becomes evident that..."
"This indicates that..."	"This observation suggests that..."

"The experimental findings can be summarized as follows..."	"In summary, the key findings from this experiment are..."
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Placeholder Instructions for Figures

Figure	Where to Insert	Caption
Figure 2	After Section 4.4, first mention	<i>"Figure 2: F1 Score Across Hospitals — Stability Comparison"</i>
Figure 3	After Section 4.4, second mention	<i>"Figure 3: Model Stability — F1 Spread Across Hospitals"</i>

Would you like me to now write the Discussion section, or would you like to review this Results section first?